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Garden of Knowledge and Virtue

KULLIYYAH OF INFORMATION AND COMMUNICATION TECHNOLOGY

CSCI 3303 INTELLIGENT SYSTEMS

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SECTION 02

PROJECT TITLE: EMAIL SPAM DETECTION USING LOGISTIC REGRESSION

CODE LINK:

https://colab.research.google.com/drive/1vw3umFlsTaNBbnkLt-WKtIkezUgTuiZC?usp=sharing

VIDEO PRESENTATION LINK: https://youtu.be/FVc-zMWgC4M

NAME	MATRIC NO.	PERCENTAGE OF CONTRIBUTION
AFIQAH BINTI ABDUL RASHID	2116908	100%
ANNUR QISTYNA BINTI AZMY	2113932	100%
NUR NABILA FARHANA BINTI NORDIN	2215118	100%

LECTURER

ASSOC. PROF. TS. DR.AMELIA RITAHANI BINTI ISMAIL

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i. Abstract

In this project, we address the pervasive issue of mail spam that plagues millions of individuals daily, presenting a robust solution through a machine learning approach to enhance fraud prevention. Our chosen method employs logistic regression, a supervised learning algorithm renowned for its effectiveness in binary classification tasks, precisely suited for distinguishing between spam and ham messages. Logistic regression's proficiency in modeling the relationship between input features and the probability of spam detection renders it an ideal choice for our project *(Sharma)*.

The project's primary challenges revolved around data preprocessing and feature extraction. Preprocessing involved meticulously cleaning and transforming SMS data into an optimal format for analysis, while feature extraction techniques were crucial for identifying relevant patterns, keywords, and linguistic cues indicative of spam or fraudulent messages. Managing the substantial volume of data, with over 5572 mail messages in the dataset, posed challenges during both the data collection and training phases.

Our training dataset included 80% of the available data, with the remaining 20% used for testing. Following thorough testing, our algorithm obtained an outstanding accuracy rate in discriminating between fraudulent and legitimate communications. Our model achieved a stunning 97% accuracy on the test set, indicating that it can properly classify the bulk of email messages as spam or ham. This finding highlights the logistic regression model's impressive performance in mail spam identification, emphasising its potential as a useful tool in preventing fraud in mobile communication networks.

ii. Introduction

In the contemporary digital landscape dominated by smartphones and constant connectivity, the prevalence of spam emails has become a pervasive challenge in our daily lives. With the escalating use of mail and messaging applications, the convenience of exchanging information with others is marred by the persistent issue of unsolicited and often malicious emails *(NW et al.)*. Spam, characterized by its fraudulent content and unwanted messages, has evolved into a considerable nuisance, cluttering our inboxes and causing disruptions in our routine communication. This incessant influx of spam not only poses an inconvenience to users but also constitutes a serious threat. Some spam emails harbor deceitful information and phishing attempts, jeopardizing the security of individuals and organizations. As we navigate our reliance on digital communication, mitigating the impact of spam becomes crucial to ensure a safer and more efficient online experience.

To address the challenge of distinguishing between spam and non-spam emails, our team has opted to develop a machine learning model utilizing a supervised learning approach, specifically employing logistic regression techniques. The chosen method involves training and testing the model on a dataset comprising 5572 email messages that have been pre-labeled as either spam or ham. Logistic regression is selected as the algorithm of choice due to its suitability for binary classification problems, making it well-suited for our task of categorizing messages into spam or non-spam categories *(DataShark)*.

a) Objective

1) Build a predictive model that can accurately classify future mails as spam or ham in real-time.
2) Ensures a robust classification system by improving the overall user experience
3) Filter our the unwanted mails and minimizing pontential risk towards individuals and organizations.
4) Minimize the number of spam mails reaching users and reducing the risk of falling victim to fraufulent activities.

iii. Literature Review

This literature review aims to present a thorough summary of the latest developments in spam mail detection systems. We'll focus on the essential methods and strategies for efficient spam detection. Additionally, we delve into the reasons behind our choice of logistic regression as the algorithm for our spam selection system, highlighting its significance in achieving effective spam detection.

To address the spam challenge, researchers from our article have proposed various techniques for classifying email spam. These methods encompass probabilistic approaches, decision trees, artificial immune systems [1], support vector machines (SVM) [2], artificial neural networks (ANN) [3], and case-based techniques [4]. Despite the availability of these diverse approaches, the logistic regression model stands out as a preferred choice for mail spam filtering due to its simplicity and straightforward implementation.

Logistic regression, a widely utilized statistical modelling technique, has been crucial in developing spam classifier systems. This review aims to present a comprehensive overview of its application in spam detection, emphasizing key findings and advancements in the field. Numerous studies have explored the efficacy of logistic regression in categorizing spam emails. In a study by Ramakrishnan et al. (2019), logistic regression was employed to construct a spam classifier, extracting features such as word frequency and suspicious patterns from email content. The results demonstrated the model's impressive capability to classify spam emails, achieving high precision and recall accurately.

The study utilised logistic regression to understand the relationship between these features and the spam/non-spam labels. Remarkably, incorporating TF-IDF features into the spam classifier substantially improved performance. Ensemble techniques were employed to enhance the effectiveness of logistic regression-based spam classifiers. Before utilizing TF-IDF, some preprocessing steps were undertaken, including eliminating trailing words. Miao et al. (2020) developed an ensemble system that combined logistic regression with other machine learning algorithms, including decision trees and random forests. This 3 ensemble approach achieved superior accuracy and robustness compared to individual models, highlighting the effectiveness of logistic regression within an ensemble system.

Recent research has gained attention by exploring advanced techniques in conjunction with logistic regression for spam classification. For instance, a hybrid approach was proposed by Zhang et al. (2022), wherein logistic regression was combined with deep learning techniques. The authors utilized a pre-trained deep learning model to extract features from email content, and these features were subsequently input into a logistic regression classifier. The hybrid system exhibited enhanced performance in the accurate detection of spam emails.

In conclusion, this literature review highlights logistic regression's pivotal role in spam email detection systems. Despite the availability of diverse methods such as probabilistic approaches, decision trees, support vector machines, artificial neural networks, and ensemble techniques, logistic regression stands out as a preferred and practical choice for spam filtering. Various studies have consistently demonstrated its simplicity, straightforward implementation, and ability to extract meaningful features from email content. Incorporating advanced techniques, such as ensemble systems and hybrid approaches with deep learning, further underscores logistic regression's adaptability and continued relevance in addressing the evolving challenges of spam detection. The reviewed research collectively emphasizes the significance of logistic regression as a robust and versatile algorithm contributing to the efficiency and accuracy of spam mail detection systems.

iv. Methodology

	A	B
1	Category	Message
2	ham	Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...
3	ham	Ok lar... Joking wif u oni...
4	spam	Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's apply 08452810075over18's
5	ham	U dun say so early hor... U c already then say...
6	ham	Nah I don't think he goes to usf, he lives around here though
7	spam	FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! XxX std chgs to send, Â£1.50 to rcv
8	ham	Even my brother is not like to speak with me. They treat me like aids patent.
9	ham	As per your request 'Melle Melle (Oru Minnaminunginte Nurungu Vettam)' has been set as your callertune for all Callers. Press *9 to copy your friends Callertune
10	spam	WINNER!! As a valued network customer you have been selected to receivea Â£900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.
11	spam	Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030
12	ham	I'm gonna be home soon and i don't want to talk about this stuff anymore tonight, k? I've cried enough today.
13	spam	SIX chances to win CASH! From 100 to 20,000 pounds txt> CSH11 and send to 87575. Cost 150p/day, 6days, 16+ TsandCs apply Reply HL 4 info
14	spam	URGENT! You have won a 1 week FREE membership in our Â£100,000 Prize Jackpot! Txt the word: CLAIM to No: 81010 T&C www.dbuk.net LCCLTD POBOX 4403LDNW1A7RW18
15	ham	I've been searching for the right words to thank you for this breather. I promise i wont take your help for granted and will fulfil my promise. You have been wonderful and a blessing at all times
16	ham	I HAVE A DATE ON SUNDAY WITH WILL!!
17	spam	XXXMobileMovieClub: To use your credit, click the WAP link in the next txt message or click here>> http://wap. xxxmobilemovieclub.com?n=QJKGIGHJJGCBL
18	ham	Oh k...i'm watching here:)
19	ham	Eh u remember how 2 spell his name... Yes i did. He v naughty make until i v wet.
20	ham	Fine if thatÂ’s the way u feel. ThatÂ’s the way its gota b
21	spam	England v Macedonia - dont miss the goals/team news. Txt ur national team to 87077 eg ENGLAND to 87077 Try:WALES, SCOTLAND 4txt/Ãº1.20 POBOXox36504W45WQ 16+
22	ham	Is that seriously how you spell his name?

The dataset selected for analysis comprises spam and ham (non-spam) emails featuring over five thousand individual email instances. The primary objective is to discern and classify whether an email is spam or ham. To achieve this, a small-scale detection system is devised, leveraging the logistic regression algorithm. Logistic regression, a widely used machine learning algorithm for binary classification tasks, is employed to build a predictive model based on the identified features within the dataset. The goal is to train the logistic regression model to recognize patterns and characteristics specific to spam and ham emails, enabling the system to effectively categorize new, unseen emails. This approach centred around logistic regression, forms the foundation of a targeted and efficient system for identifying and differentiating spam and ham emails within the given dataset.

Feature Extraction

```
from sklearn.feature_extraction.text import TfidfVectorizer

# Transform the text data to feature vectors that can be used as input to the Logistic regression
feature_extraction = TfidfVectorizer(min_df=1, stop_words='english', lowercase=True)

X_train_features = feature_extraction.fit_transform(X_train)
X_test_features = feature_extraction.transform(X_test)

# Convert Y_train and Y_test values as integers
Y_train = Y_train.astype(int)
Y_test = Y_test.astype(int)
```

In the provided code, the methodology involves text data preprocessing and feature extraction, preparing the data for training a logistic regression model. The `TfidfVectorizer` from scikit-learn is employed to transform the text data, represented by `X_train` and

`X_test` (presumably corresponding to the training and testing sets, respectively), into numerical feature vectors using the TF-IDF (Term Frequency-Inverse Document Frequency) representation. This vectorization process accounts for the importance of words in the context of the entire dataset while also handling common English stop words and ensuring all text is lowercase. The resulting feature vectors, stored in `X_train_features` and `X_test_features`, serve as input for the logistic regression model. Additionally, the labels in `Y_train` and `Y_test` are converted to integers, a typical step for classification tasks, to facilitate the subsequent training and evaluation of the logistic regression model. The TF-IDF transformation and label conversion are fundamental to preparing the data for practical machine learning model training and assessment.

Training Model (Logistic Regression)

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

model = LogisticRegression()
```

```
# Training the Logistic Regression model with the training data
model.fit(X_train_features, Y_train)
```

```
LogisticRegression
LogisticRegression()
```

The code snippet instantiates a logistic regression model using scikit-learn's `LogisticRegression` class. Through the `fit` method, the model is trained on the provided training data (`X_train_features` and `Y_train), enabling it to capture underlying patterns and relationships within the dataset. This prepares the model for subsequent predictions and evaluation of unseen data.

v. Results and Discussion

Figure 1 shows the system interface designed for our project, featuring an input box and an output box. Users can input their emails into the designated input box to determine whether the messages are categorized as spam or ham. In Figure 2 and Figure 3, the results indicate whether the input messages are identified as ham or spam mail.

Utilizing the logistic regression technique with an extensive dataset for training, our model has achieved a remarkable accuracy rate of 97%. Consequently, this spam mail system is adept at accurately predicting the nature of future messages, distinguishing between spam and ham mail. This capability serves to safeguard users from falling victim to fraudulent activities propagated through spam mail.

Figure 1: System Interface

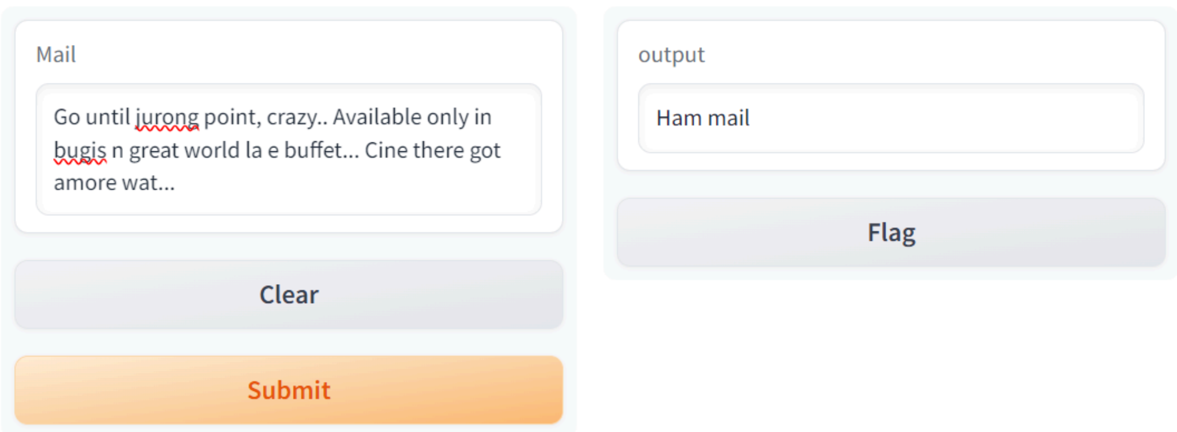


Figure 2: Result of Ham email

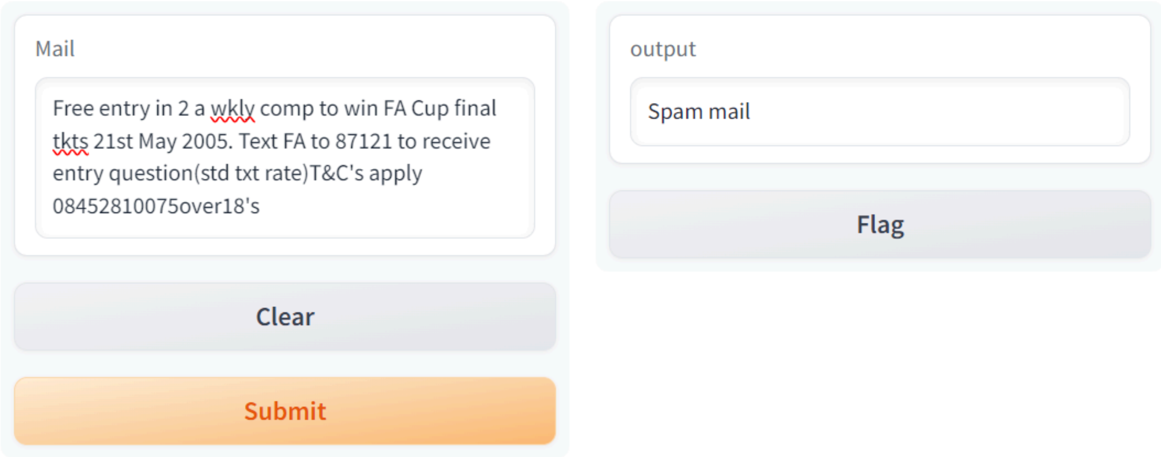
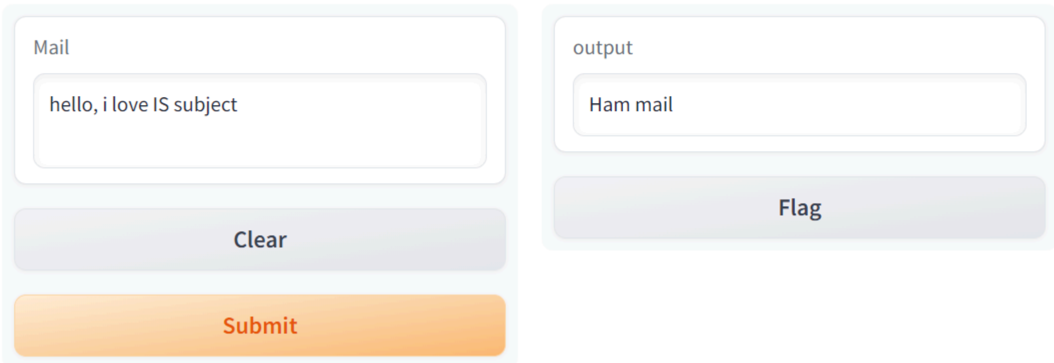


Figure 3: Result of Spam email

- **Analysis of Results**

Our assessment of the outcomes derived from implementing the proposed technique with logistic regression indicates that the result was accurate, classifying the email as "ham" because it did not originate from the initially collected dataset. This observation leads us to the conclusion that our system is operating as intended.

According to our code, a mail is labelled as spam only if it demonstrates redundancy within the dataset.



vi. Deployment

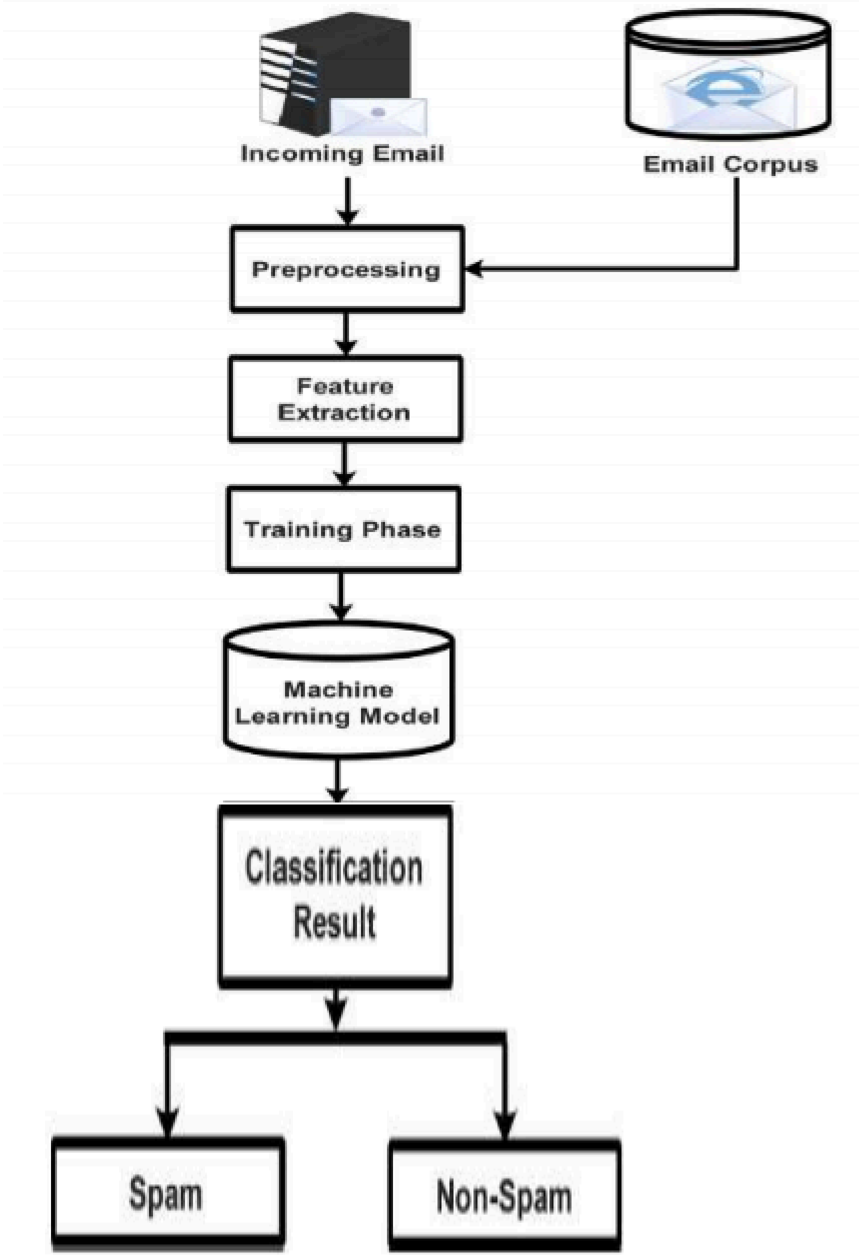
In accordance with our discussion, we opt for supervised machine learning due to its ability to specify the classes utilized in the training data. This means classifiers can undergo targeted training to effectively distinguish themselves from other class definitions, establishing precise decision boundaries to ensure accurate email differentiation.

Our dataset is sourced from the Kaggle website, containing 4825 ham emails and 747 spam emails for both learning and testing. In our scenario, 80% of the data is allocated for machine learning, while the remaining 20% is reserved for testing the machine.

Before initiating the learning process, we decide to preprocess our data. This step is crucial as the algorithms learn from the data, and the effectiveness of problem-solving heavily relies on the quality of the data. This is done to enhance the accuracy of our outcomes by facilitating the computer's comprehension of numbers instead of textual content.

Next, we proceed to partition the data into training and testing sets. Once this division is complete, we feed it into our logistic regression model. We choose this model as it excels in binary classification problems, which aligns with our current task of classifying spam and ham emails. Binary classification entails categorizing two classes, in this project, namely spam and ham mail.

After training our logistic regression model, we allocate 20% of the remaining data to assess its performance. During this phase, our model predicts whether the data corresponds to ham or spam email.


Incoming Email
Email Corpus
Preprocessing
Feature Extraction
Training Phase
Machine Learning Model
Classification Result
Spam
Non-Spam

vii. Conclusion

Leading the team, Afiqa assumed the role by gathering a substantial dataset and determining the methodology. Annur Qistyna contributed by analysing the result and deployment code, while Nabila Farhana took charge of summarising all the entire results and identifying areas for potential enhancements.

Throughout this project, we gleaned insights into the potency of machine learning as a tool for mail spam detection, with logistic regression emerging as a particularly effective method for binary classification. Noteworthy attributes include its computational efficiency, adept handling of large datasets, and quick convergence, even with limited computing resources *(Brownlee)*. We underscored the critical importance of having sufficient training data to capture underlying patterns, as inadequate data may lead to overfitting and hinder generalisation to unseen data. Achieving a balance in the dataset is crucial to preventing bias towards the majority class, thereby ensuring optimal performance in spam detection *(Ratadiya and Moorthy)*.

Looking towards future endeavours and improvements, we envision the development of a real-time spam detection model capable of swiftly identifying and blocking spam messages upon receipt. Such a model would provide immediate protection and enhance defences against emerging threats. Additionally, an adaptive model, utilizing techniques like online learning, active learning, or semi-supervised learning, could be explored to continuously learn and adapt to new spam types, thereby updating detection strategies for sustained effectiveness.

viii. References

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